

AAE 507: Principles of Dynamics

Problem Set 1 & 2

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PS1 Due: Sep 4, 2026 at 11:59 PM ET

PS2 Due: Sep 11, 2026 at 11:59 PM ET
(submission at Brightspace>Content>Gradescope)

Use 1×10^{-12} for the integration tolerance of ordinary differential equations unless otherwise requested. When plotting trajectories in a 2-D or 3-D space, use equal axes for the x -, y -, and z - axes.

Problem Set 1 Review: Particle Kinematics and Dynamics

(1.1) Particle Kinematics

- (a): [**AI use Level: 0**] Let $\mathbf{a}$ be a physical vector, and let ${}^{\mathcal{N}}[\mathbf{a}]$ and ${}^{\mathcal{B}}[\mathbf{a}]$ denote its matrix-format (column-vector) representations in two different coordinate systems $\mathcal{N}$ and $\mathcal{B}$.
- (a.1): **Explain** why ${}^{\mathcal{N}}[\mathbf{a}] \neq {}^{\mathcal{B}}[\mathbf{a}]$ in general, even though both represent the same physical vector $\mathbf{a}$.
- (a.2): A classmate writes $\dot{\mathbf{r}}$ to mean “the time derivative of $\mathbf{r}$ ” without specifying an observing frame. **Explain** why this is ambiguous, and **state** the convention this course uses for what $\dot{\mathbf{r}}$ means.
- (b): [**AI use Level: 0**] $\mathcal{B}$ -frame with basis vectors $\{\hat{\mathbf{b}}_1, \hat{\mathbf{b}}_2, \hat{\mathbf{b}}_3\}$ rotates relative to the inertial frame $\mathcal{N} : \{\hat{\mathbf{n}}_1, \hat{\mathbf{n}}_2, \hat{\mathbf{n}}_3\}$ with angular velocity $\boldsymbol{\omega}_{\mathcal{B}/\mathcal{N}} = \omega_0 \hat{\mathbf{b}}_3$, where ω_0 is constant. $\mathcal{B}$ -frame and $\mathcal{N}$ -frame are aligned at $t = 0$.
- (b.1): There is a point whose location is given by $\mathbf{r} = r_0 \hat{\mathbf{b}}_1$, where r_0 is constant. **Derive** $\dot{\mathbf{r}}$ in vector format first, and then **express** it in $\mathcal{B}$ -frame and $\mathcal{N}$ -frame as a function of t .
- (b.2): Now suppose the point is moving w.r.t. $\mathcal{B}$ -frame at constant rate, expressed as $\mathbf{r}(t) = r_0 \hat{\mathbf{b}}_1 + \dot{r}_0 t \hat{\mathbf{b}}_1$, where $\dot{r}_0$ is constant. **Derive** $\dot{\mathbf{r}}$ in vector format, and **discuss** which term is new relative to the case with $\mathbf{r} = r_0 \hat{\mathbf{b}}_1$ and physically why it appears.
- (c): [**AI use Level: 0**] In lecture we derived velocity and acceleration in *spherical* coordinates. Here we consider *cylindrical* coordinates. Let $\mathcal{C} : \{\hat{\mathbf{e}}_\rho, \hat{\mathbf{e}}_\phi, \hat{\mathbf{e}}_z\}$, where ρ is the radial distance from a fixed z -axis, ϕ is the azimuth about $\hat{\mathbf{e}}_z$ measured from a fixed reference direction $\hat{\mathbf{e}}_x$, and z is the axial coordinate.
- (c.1): **Draw** a sketch showing $\mathbf{r}$ (position of a point) and $\{\hat{\mathbf{e}}_\rho, \hat{\mathbf{e}}_\phi, \hat{\mathbf{e}}_z\}$. **Express** $\mathbf{r}$ in terms of $\{\hat{\mathbf{e}}_\rho, \hat{\mathbf{e}}_\phi, \hat{\mathbf{e}}_z\}$.
- (c.2): **Determine** the value of $\boldsymbol{\omega}_{\mathcal{C}/\mathcal{N}}$, $\mathbf{v}(=\dot{\mathbf{r}})$, and $\mathbf{a}(=\ddot{\mathbf{r}})$ in terms of $\{\hat{\mathbf{e}}_\rho, \hat{\mathbf{e}}_\phi, \hat{\mathbf{e}}_z\}$.
- (c.3): **Confirm** that $\hat{\mathbf{e}}_\rho, \hat{\mathbf{e}}_\phi$ components of your $\mathbf{a}$ reduce to the planar polar-coordinate result ($\ddot{\rho} - \rho\dot{\phi}^2$ and $\rho\ddot{\phi} + 2\dot{\rho}\dot{\phi}$, respectively) when $z = 0$.

(1.2) Orbital Dynamics

- (d): [**AI use Level: 0**] Using $\ddot{\mathbf{r}} = -\mu\mathbf{r}/r^3$, **prove** that the specific angular momentum $\mathbf{h} = \mathbf{r} \times \mathbf{v}$ and the eccentricity vector $\mathbf{e} = \frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r}$ are both constant.

Table 1: Reference orbit

Constants			Classical orbital elements at epoch			
μ_{earth}	R_{earth}	J_2	a	e	i	$\Omega_0, \omega_0, \nu_0$
398600 km ³ /s ²	6378.137 km	1.08263×10^{-3}	6978 km	0.01	45 deg	0 deg

[Hint] For $\dot{\mathbf{h}} = \mathbf{0}$: differentiate the definition of $\mathbf{h}$. For $\dot{\mathbf{e}} = \mathbf{0}$: differentiate the definition of $\mathbf{e}$, substitute $\ddot{\mathbf{r}} = -\mu\mathbf{r}/r^3$, and use the vector identity $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$.

- (e): **[AI use Level: 0]** As discussed in class, Kepler's equation $M = E - e \sin E$ does not have a closed-form solution and requires an iterative method to obtain E for given e, M . Let us solve the Kepler's equation using Newton's method.

Newton's method

Newton's method finds a root of a scalar equation $f(x) = 0$ by using a gradient of f at the current guess. Mathematically, linearizing f about x_k gives $f(x) \approx f(x_k) + \nabla_x f(x_k)(x - x_k)$, and solving this linear equation for its zero yields the update rule: $x_{k+1} = x_k - f(x_k)/\nabla_x f(x_k)$. Thus, numerical implementation of Newton's method is: (i) initialize $x \leftarrow x_0$, (ii) compute $f(x)$ and $\nabla_x f(x)$, (iii) form $\Delta x = -f(x)/\nabla_x f(x)$, (iv) update $x \leftarrow x + \Delta x$, and (v) repeat (ii)-(iv) until $|\Delta x|$ is sufficiently small.

- (e.1): The scalar equation we need to solve is $f(E) = E - e \sin E - M = 0$. **Verify** that the update rule for Newton's method is $E_{k+1} = E_k - f(E_k)/(1 - e \cos E_k)$.
- (e.2): **Implement** Newton's method for Kepler's equation, and **report** the code snippet you implemented.
- (e.3): Assuming a satellite is in orbit with $e = 0.4$ and $M = 60$ deg, **solve** Kepler's equation using Newton's method. Use $E_0 = M$ to initialize. Iterate until $|\Delta E_k|$ is below 10^{-8} rad. **Report** the values of E_k , $f(E_k)$, $\nabla_E f(E_k)$, and ΔE_k for *every* iteration k .
- (f): **[AI use Level: 0]** Let us calculate and compare the magnitude of different perturbations.
- (f.1): **Report** the Cartesian state vector at the epoch $t = 0$ using the orbit information given in Table 1.
- (f.2): **Report** the magnitude of each of the four perturbing accelerations below at the epoch.
- J_2 : use Eq. (1), where $\mathbf{r} = (x, y, z)$ and $r_o = R_{\text{earth}}$.

$$\mathbf{a}_{J2} = -\frac{3\mu J_2 r_o^2}{2r^5} \left[\left(1 - 5\frac{z^2}{r^2}\right) x \hat{\mathbf{n}}_1 + \left(1 - 5\frac{z^2}{r^2}\right) y \hat{\mathbf{n}}_2 + \left(3 - 5\frac{z^2}{r^2}\right) z \hat{\mathbf{n}}_3 \right] \quad (1)$$

- Third body (Sun and Moon, computed separately): See the lecture slides. Use $\mu_{\text{Moon}} = 4902.8$ km³/s², $d_{\text{Moon}} = 384,400$ km, $\mu_{\text{Sun}} = 1.327 \times 10^{11}$ km³/s², $d_{\text{Sun}} = 1.496 \times 10^8$ km. Note: You have some flexibility to define the Sun and Moon directions at the epoch. **Report** the directions you assumed.
- SRP: use the cannon-ball model from the slides. Use $A/m = 0.01$ m²/kg and $C_R = 1.3$, which is representative of some small satellites. **Be careful about the unit.**
- Atmospheric drag: Use a simplified model (**Be careful about the unit**)

$$\mathbf{a}_{\text{drag}} = -\frac{1}{2} C_D \frac{A}{m} \rho v^2 \hat{\mathbf{v}}, \quad v := \|\mathbf{v}\|_2, \quad \hat{\mathbf{v}} := \frac{\mathbf{v}}{v} \quad (2)$$

with $C_D = 2.2$, $A/m = 0.01$ m²/kg (same as SRP), and a representative atmospheric density at 600 km altitude, $\rho = 1.454 \times 10^{-13}$ kg/m³.

- (f.3): **Discuss** how these perturbations compare (what is the largest/smallest, how much spread they have, why does it make sense?, etc)
- (g): [**AI use Level: 1**] Using the Cartesian state from above as the initial condition, numerically **propagate** the unperturbed two-body EoMs for at least 10 orbits.
- (g.1): **Plot** the resulting trajectory in 3D.
- (g.2): **Compute** the specific energy $\mathcal{E}(t)$, specific angular momentum $\mathbf{h}(t)$, and eccentricity vector $\mathbf{e}(t)$, and **plot** their time histories. **Discuss** the results (does it make sense? why?).
- (g.3): Repeat (g.1)-(g.2), but now add the J_2 acceleration to the EoMs. **Plot** the resulting trajectory, and **compare** it against the unperturbed result. **Discuss** how the two differ.

(1.3) Optional (extra credit)

- (a): [**AI use Level: 0**] Starting from the spherical-harmonics gravitational potential given in the lecture slides, specialize to the zonal ($m = 0$), $\ell = 2$ term only, using $C_{20} = -J_2$ and the Legendre polynomial $P_{20}(\sin \delta) = \frac{1}{2}(3 \sin^2 \delta - 1)$ with $\sin \delta = z/r$, to show that

$$U_{J_2} = \frac{\mu J_2 r_o^2}{2r^5} (3z^2 - r^2).$$

Then, **calculate** $\mathbf{a}_{J_2} = -\nabla U$ component-by-component to reproduce Eq. (1).

- (b): [**AI use Level: 1**] **Implement** the J_2 -perturbed propagation from Problem (g) using the *spherical*-coordinate EoMs instead of Cartesian. Convert your spherical-coordinate result back to Cartesian position $\mathbf{r}(t)$ and **compare** it against your original Cartesian propagation. **Discuss** any practical differences between the two versions (e.g., step-size behavior, ease of adding perturbations, etc).
- (c): [**AI use Level: 1**] Select another perturbation in addition to J_2 and propagate the EoMs at several semi-major axis values, including 6500, 6800, 7100, and 7400 km. At each altitude, **run** the propagation twice, once with J_2 only and once with J_2 plus your chosen perturbation, and **discuss** how the two diverge, and how that divergence changes across the altitude sweep.

Note: atmospheric density depends on altitude. If you choose drag, use a simplified exponential model

$$\rho(r) \approx \rho_0 \exp\left(-\frac{r - 6978 \text{ km}}{H}\right), \quad \rho_0 = 1.454 \times 10^{-13} \text{ kg/m}^3, \quad H \approx 60 \text{ km}.$$

Problem Set 2 Review: Rigid-body Kinematics and Dynamics

(2.1) Rigid-Body Kinematics and Dynamics

- (a): [**AI use Level: 0**] For each of the three matrices below, **check** whether it is a valid DCM.

$$A = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0.1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

- (b): [**AI use Level: 0**] Consider a principal rotation of angle $\theta = 120$ deg about the axis $\hat{\lambda} = \frac{1}{\sqrt{3}}[1, 1, 1]^\top$.

- (b.1): **Compute** the corresponding Euler parameters $\boldsymbol{\epsilon}$. **Confirm** $\|\boldsymbol{\epsilon}\| = 1$.
- (b.2): **Compute** the corresponding DCM $[BN]$. **Verify** it is a valid DCM.
- (b.3): **Recompute** $[BN]$ using $-\boldsymbol{\epsilon}$ instead of $\boldsymbol{\epsilon}$, and **confirm** you get the *same* DCM.

- (c): [**AI use Level: 0**]

- (c.1): Starting from $\dot{\mathbf{H}}_c = \boldsymbol{\tau}_c$, **show** the Euler's rotational equations of motion: $I_c \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (I_c \boldsymbol{\omega}) = \boldsymbol{\tau}_c$. In the process, **justify** (i) why $\frac{d}{dt} \boldsymbol{\omega} = \dot{\boldsymbol{\omega}}$, and (ii) why $\frac{d}{dt} I_c = \mathbf{0}$.
- (c.2): Assuming B -frame axes are the principal axes of inertia, **derive** the three scalar differential equations from $I_c \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (I_c \boldsymbol{\omega}) = \boldsymbol{\tau}_c$.
- (d): [**AI use Level: 0**] Consider torque-free motion of an axisymmetric rigid body, $I_1 = I_2 = I_T$ and $I_3 = J$ (with $J \neq I_T$), using Euler's equations from (c).
- (d.1): **Show** that $\omega_3(t) = \omega_{3_0}$ is constant and that ω_1 and ω_2 satisfy a pair of coupled linear ODEs.
- (d.2): **Solve** the coupled linear ODEs to **show** that, with initial condition $\boldsymbol{\omega}(t_0) = [\omega_{1_0}, \omega_{2_0}, \omega_{3_0}]^\top$,

$$\omega_1(t) = \omega_{1_0} \cos(K\omega_{3_0}t) + \omega_{2_0} \sin(K\omega_{3_0}t), \quad \omega_2(t) = \omega_{2_0} \cos(K\omega_{3_0}t) - \omega_{1_0} \sin(K\omega_{3_0}t) \quad (3)$$

where $K = (I_T - J)/I_T$.

(2.2) Spacecraft Attitude Dynamics

- (e): [**AI use Level: 0**] Starting from the (approximate) gravity-gradient torque about the center of mass $\boldsymbol{\tau}_G = \frac{3\mu}{r_c^5} \mathbf{r}_c \times (I_c \mathbf{r}_c)$, **derive** the component form in B -frame

$$\tau_1 = \frac{3\mu}{r_c^5} (I_3 - I_2) r_{c3} r_{c2}, \quad \tau_2 = \frac{3\mu}{r_c^5} (I_1 - I_3) r_{c1} r_{c3}, \quad \tau_3 = \frac{3\mu}{r_c^5} (I_2 - I_1) r_{c2} r_{c1}$$

where we assume the body-frame axes are the principal axes of inertia, and ${}^B \mathbf{r}_c = [r_{c1} \ r_{c2} \ r_{c3}]^\top$.

- (f): [**AI use Level: 0**] Consider an axisymmetric rigid body with $I_1 = I_2 = I_T = 10 \text{ kg}\cdot\text{m}^2$, $I_3 = J = 5 \text{ kg}\cdot\text{m}^2$, in torque-free motion with initial angular velocity $\boldsymbol{\omega}_0 = [0.2, 0.1, 1.0]^\top \text{ rad/s}$.
- (f.1): **Numerically integrate** Euler's equations of motion for at least 120 sec. **Report** relevant plots.
- (f.2): **Compute** the value of $\boldsymbol{\omega}(t)$ using the analytical solution from Eq. (3). **Overlay** the numerical and analytical solutions. The two should agree closely; **discuss** any discrepancy you observe.

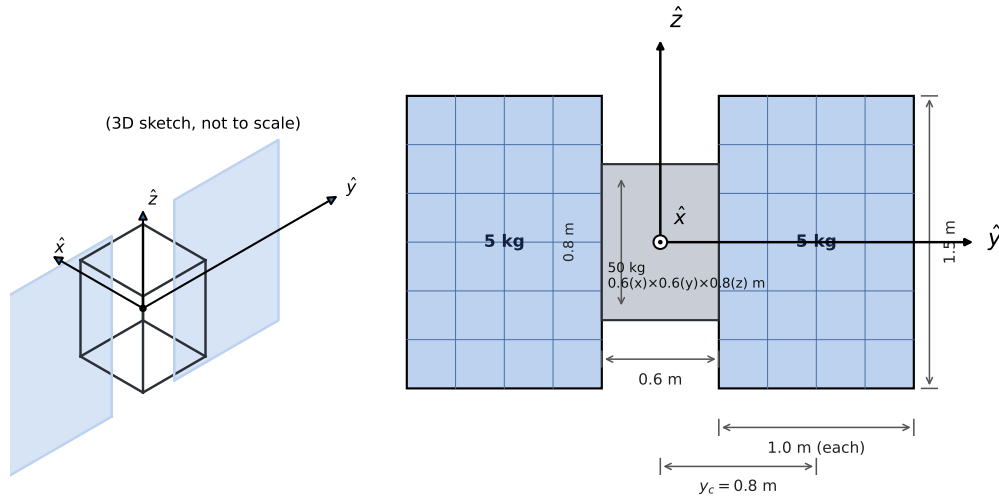


Figure 1: Bus-plus-panels satellite configuration

- (g): [**AI use Level: 0**] Consider a satellite made of a bus plus two identical solar panels, see Fig. 1.
- Bus: mass 50 kg, size $0.6 \text{ m}(\hat{x}) \times 0.6 \text{ m}(\hat{y}) \times 0.8 \text{ m}(\hat{z})$, centered at the B -frame origin.
 - Panels: two identical rectangular panels, 1.0 m (along $\hat{y}$) $\times$ 1.5 m (along $\hat{z}$), negligible thickness, 5 kg each, mounted symmetrically with their inner edge at $y = 0.3 \text{ m}$ from the origin.

- You can use the following equations without derivation. For a box-shaped object of mass m with side lengths L_x, L_y, L_z , the moment of inertia about its own center of mass (CoM) is given by

$$I_{xx} = \frac{m}{12}(L_y^2 + L_z^2), \quad I_{yy} = \frac{m}{12}(L_x^2 + L_z^2), \quad I_{zz} = \frac{m}{12}(L_x^2 + L_y^2)$$

Parallel Axis Theorem

For a rigid body of mass m with inertia tensor I_c about its own center of mass, its inertia tensor about a parallel reference point P displaced from the CoM by $\mathbf{d} \in \mathbb{R}^3$ is

$$I_P = I_c + m (\|\mathbf{d}\|^2 I_3 - \mathbf{d} \mathbf{d}^\top)$$

For a composite body, sum each component's own-centroid inertia tensor plus its own parallel-axis correction to assemble the inertia tensor about one common point.

- (g.1): **Explain** why the system's CoM is exactly at the B -frame origin.
- (g.2): **Compute** the inertia tensor of each component about the origin.
Hint: for panels, use the parallel axis theorem to shift the inertia tensors to the system origin.
- (g.3): **Compute** the assembled inertia tensor I_c about the system origin by summing the three contributions. **Verify** that every product of inertia (I_{xy}, I_{xz}, I_{yz}) is exactly zero.
- (h): [**AI use Level: 1**] Consider the satellite in Fig. 1 in a *circular* ($e = 0$) version of the orbit in Table 1.
- (h.1): **Derive** the attitude equations of motion: Euler's equations with the gravity-gradient torque, together with a kinematic differential equation (KDEs) for your choice of attitude representation (DCM, Euler parameters, or Euler angles).
Hint: Since the orbit is circular, the orbit radius is constant. This fact simplifies the EoMs and numerical integration.
- (h.2): **Choose** the initial attitude to be “nadir-pointing,” where one body axis is aligned with $-\mathbf{r}_c$, while another aligned with the orbit-normal direction $\mathbf{r}_c \times \mathbf{v}_c$. You have some freedom in how you define “nadir-pointing”; **state** your convention and **report** the attitude initial condition.
- (h.3): Perturb your initial attitude by a small offset (a few degrees) with a small initial body rate (**report** the values), **integrate** the attitude motion for several orbits, and **plot** the resulting attitude/rate time history. **Describe** what you observe empirically (does the small initial perturbation stay bounded, grow, or oscillate?)

(2.3) Optional (extra credit)

- (a): [**AI use Level: 1**] Consider the nadir-pointing satellite attitude simulation under the gravity-gradient torque above, and analyze its linear stability.
- (a.1): **Identify** the nadir-pointing equilibrium between the body and the orbit frame under the gravity-gradient torque.
- (a.2): **Introduce** a perturbation about this equilibrium, and **linearize** EoMs about the equilibrium.
- (a.3): **Derive** its eigenvalues in terms of $K_1 = (I_2 - I_3)/I_1$, $K_2 = (I_3 - I_1)/I_2$, $K_3 = (I_1 - I_2)/I_3$.
- (a.4): **Evaluate** your eigenvalues numerically using the actual values from (h), and **determine** whether the equilibrium is linearly stable. **Discuss** its consistency with your empirical observation in (h.3).
- (b): [**AI use Level: 1**] Instead of the circular orbit used in (h), consider the gravity-gradient attitude simulation with the J_2 -perturbed orbital motion $\mathbf{r}_c(t)$ that you propagated in [Problem Set 1](#) with $e = 0.01$. With this setting, **numerically integrate** the gravity-gradient attitude motion by using the numerically propagated $\mathbf{r}_c(t)$ (interpolate as needed; **deval** becomes handy here). **Compare** the resulting attitude motion against your circular-orbit result, and **discuss**.